

Hansol Kang

Ph.D. Candidate
Sungkyunkwan University
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EDUCATION

Mar. 2017 ~ Present	Sungkyunkwan University Department of Mechanical Engineering Thesis: 구동기 식별을 통한 고감속기 사족보행 로봇의 강인한 보행 및 전복 복원 학습 <i>Advisor: Hyouk Ryeol Choi</i> <i>Ph.D. Student</i> GPA: 3.86 / 4.5	Suwon, Korea
Mar. 2012 ~ Feb. 2017	Ajou University Department of Mechanical Engineering <i>B.S. in Mechanical Engineering</i> GPA: 4.0 / 4.5	Suwon, Korea

PUBLICATIONS (SCIE/ESCI)

1. Luong Tin Phan, Yoon Haeng Lee, Young Hun Lee, Hyun-Yong Lee, **Hansol Kang**, Hyouk Ryeol Choi, "Study on effects of spinal joint for running quadruped robots", *Intelligent Service Robotics*, (2020)
2. Yoon Haeng Lee, Young Hun Lee, Hyunyong Lee, **Hansol Kang**, Jun Hyuk Lee, Luong Tin Phan, Sungmoon Jin, Yong Bum Kim, Dong-Yeop Seok, Seung Yeon Lee, Hyungpil Moon, Ja Choon Koo, Hyouk Ryeol Choi, "Development of A Quadruped Robot System with Torque-Controllable Modular Actuator Unit", *IEEE Transactions on Industrial Electronics*, (2020)
3. Young Hun Lee, Yoon Haeng Lee, Hyunyong Lee, **Hansol Kang**, Jun Hyuk Lee, Ji Man Park, Yong Bum Kim, Hyungpil Moon, Ja Choon Koo, Hyouk Ryeol Choi, "Whole-body control and angular momentum regulation using torque sensors for quadrupedal robots", *Journal of Intelligent & Robotic Systems*, (2021)
4. Young Hun Lee, Hyunyong Lee, **Hansol Kang**, Jun Hyuk Lee, Ji Man Park, Chunghyuk Kang, Yoon Haeng Lee, Yong Bum Kim, Hyouk Ryeol Choi, "Balance Recovery Based on Whole-Body Control Using Joint Torque Feedback for Quadrupedal Robots", *Journal of Mechanisms and Robotics*, (2021)
5. Yeong Gwang Son, Tat Hieu Bui, Juyong Hong, Yong Hyeon Kim, Seung Jae Moon, Chun Soo Kim, Issac Rhee, **Hansol Kang**, Hyouk Ryeol Choi, "CoAS-Net: Context-Aware Suction Network With a Large-Scale Domain Randomized Synthetic Dataset", *IEEE Robotics and Automation Letters*, (2023)
6. **Han Sol Kang**, Hyun Yong Lee, Ji Man Park, Seong Won Nam, Yeong Woo Son, Bum Su Yi, Jae Young Oh, Jun Ha Song, Soo Yeon Choi, Bo Geun Kim, Hyun Seok Kim, Hyouk Ryeol Choi, "ARS: AI-Driven Recovery Controller for Quadruped Robot Using Single-Network Model",

7. Soo Yeon Choi,Hyunyong Lee, Ji Man Park, **Hansol Kang**, Seong Won Nam, Bum Su Yi, Yeong Woo Son, Jae Young Oh, Jun Ha Song, Bo Geun Kim, Dae Geun Song, Hyouk Ryeol Choi, "Development of Two-DOF Simultaneous Gravity Compensator for Mobile Manipulator", *IEEE Robotics and Automation Letters*, (2025)
8. Yeong Woo Son,Hyunyong Lee, **Hansol Kang**, Jiman Park, Seongwon Nam, Jaeyoung Oh, Bumsu Yi, Junha Song, Bogeun Kim, Daegeun Song, Hyouk Ryeol Choi, "RoA-Planner: Rotatable Area based Path Planner in Dense Spaces", *IEEE Transactions on Automation Science and Engineering*, (2025)
9. Yeongwoo Son,Hyunyong Lee, **Hansol Kang**, Jiman Park, Seongwon Nam, Jaeyoung Oh, Bumsu Yi, Hyeonwoo Yu, Hyouk Ryeol Choi, "3D RoA-Planner: Path Planner for Quadruped Robots in Confined Spaces Using 3D Rotatable Areas", *IEEE Robotics and Automation Letters*, (2026)
10. SeongWon Nam,Hyunyong Lee, **Hansol Kang**, Jiman Park, Yeongwoo Son, Bumsu Yi, Jaeyoung Oh, Hyouk Ryeol Choi, "Real-Time Control-Constrained DDP for Underactuated Balancing of Legged Robots", *IEEE Robotics and Automation Letters*, (2026)
11. **Hansol Kang**, Hyunyong Lee, Jiman Park, Seongwon Nam, Yeongwoo Son, Bumsu Yi, Jaeyoung Oh, Hyeonwoo Yu, Hyouk Ryeol Choi, "Enhancing Sim-to-Real Transfer for a High-Gear-Ratio Quadruped Robot via Extended Actuator Dynamics Identification", *Machines*, (2026)

CONFERENCES

1. Yoon Haeng Lee, Young Hun Lee, Hyunyong Lee, Luong Tin Phan, **Hansol Kang**, Yong Bum Kim, Hyouk Ryeol Choi, "Development of torque controllable leg for running robot, AiDIN-IV.", IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2017
2. Young Hun Lee, Yoon Haeng Lee, Hyunyong Lee, Luong Tin Phan, **Hansol Kang**, Uikyum Kim, Jeongmin Jeon, Hyouk Ryeol Choi, "Trajectory design and control of quadruped robot for trotting over obstacles.", IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2017
3. Luong Tin Phan, Yoon Haeng Lee, Young Hun Lee, Hyunyong Lee, **Hansol Kang**, Hyouk Ryeol Choi, "Study on quadruped bounding with a passive compliant spine.", IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2017
4. Luong Tin Phan, Yoon Haeng Lee, Young Hun Lee, Hyun-Yong Lee, **Hansol Kang**, Hyouk Ryeol Choi, "Effects of spinal joint on quadrupedal bounding", 14th International Conference on Ubiquitous Robots and Ambient Intelligence (URAI), 2018
5. Yoon Haeng Lee, Young Hun Lee, Hyun-Yong Lee, **Hansol Kang**, Luong Tin Phan, Sungmoon Jin, Yong Bum Kim, Dong-Yeop Seok, Seung Yeon Lee, Hyungpil Moon, Ja Choon Koo, Hyouk Ryeol Choi, "Force-controllable Quadruped Robot System with Capacitive-type Joint Torque Sensor", International Conference on Robotics and Automation (ICRA), 2019
6. Seungjun Woo, Jinjae Shin, Yoon Haeng Lee, Young Hun Lee, Hyun-Yong Lee, **Hansol Kang**, Hyouk Ryeol Choi, Hyungpil Moon, "Stair-mapping with Point-cloud Data and Stair-modeling for Quadruped Robot", 16th international conference on ubiquitous robots (UR), 2019
7. Young Hun Lee, Ja Choon Koo, Hyouk Ryeol Choi, Yoon Haeng Lee, Hyunyong Lee, **Hansol Kang**, Yong Bum Kim, Jun Hyuk Lee, Luong Tin Phan, Sung Moon Jin, Hyungpil Moon, "Whole-Body Motion and Landing Force Control for Quadrupedal Stair Climbing.", IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2019
8. **Hansol Kang**, Hyunyong Lee, Jiman Park, Seongwon Nam, Yeongwoo Son, Bumsu Yi, Jaeyoung Oh, Bogeun Kim, Daegeun Song, Hyouk Ryeol Choi, "Enhancing Sim-to-Real Transfer in High Gear Ratio Quadruped Robot via Short-Term Temporal Information", International Conference on Robotics and Automation, Atlanta, United States (May. 2025) - Late Breaking Results

RESEARCH INTERESTS

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- Legged robot
 - Reinforcement learning control
 - Humanoid robot
 - Quadrupedal robot

AWARDS AND HONORS

- Dean's List, Ajou University, Korea (Jul. 2016)
- Global Ph.D. Fellowship, National Research Foundation of Korea, Korea (Mar. 2018)
- Grand Prize, 2nd Dronbot Challenge, Daejeon-si, ROK Army Training & Doctrine Command, Agency for Defense Development, Korea (Aug. 2019)
- 사족보행로봇 부문 대상, POSCO Industrial AI-Industrial Challenge, POSCO HOLDINGS, POSCO DX, Korea (Sep. 2025)
- 우수논문상, 제어로봇시스템학회, Korea (Jun. 2025)

PROJECTS

- 고속주행을 위한 생체모방 4족보행로봇의 연성이 주행성능에 미치는 영향에 관한 연구 및 실험적 검증, 성균관대학교, Korea (Mar. 2017 ~ Apr. 2017)
- 미래첨단도시 활용을 위한 멀티모달 보행로봇 기술개발, Gyeonggido Business & Science Accelerator, Korea (Jun. 2017 ~ May. 2019)
- A Development of Ultra High Degree of Freedom Quadruped Robot with No Restrictions on Moving on Roug, National Research Foundation of Korea, Korea / Principal Investigator (Mar. 2018 ~ Feb. 2021)
- 3D 지면데이터 기반 힘지 이동 제어 기술 개발, 현대로템, Korea (May. 2022 ~ Dec. 2024)